



No Symbols? No Problem!

By Max 'Libra' Kersten

Table of contents



About me



Symbols and types of code



Symbol recovery



Bulk signature generation



Case study



Methodology

About me

- Max 'Libra' Kersten ([/in/Libranalysis](#) on LinkedIn, [@Libranalysis](#) on Twitter)
 - Spoke at numerous conferences (BlackHat USA/EU/MEA/Asia, DEFCON, Botconf, and others)
- Malware analyst at Trellix' Advanced Research Center
 - Published [DotDumper](#)
- I write [blogs](#) about reverse engineering
 - Including my free [Binary Analysis Course](#)
- My tools are open-sourced on [GitHub](#)
 - Such as [AndroidProjectCreator](#) and the [Mobile Malware Mimicking Framework](#)



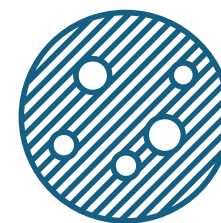
Symbols 101



Recognisable



Self-explanatory



Unambiguous

Symbols 101



Recognisable



Self-explanatory



Unambiguous

Types of code



Runtime



Library



User

Symbol recovery



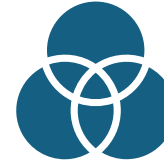
Metadata

Useful, but not always available



FunctionID

Function name and hash mapping



BSim

Used to see overlap between functions

FunctionID databases

- Automated
- Runs headless

```
INFO HEADLESS: execution starts (HeadlessAnalyzer)
INFO Opening existing project: C:\Users\malwa\fidb-test (HeadlessAnalyzer)
INFO Opening project: C:\Users\malwa\fidb-test (HeadlessProject)
INFO SCRIPT: C:\Users\malwa\ghidra_scripts\AutomaticallyCreateFunctionIdDatabases.java (HeadlessAnalyzer)
INFO AutomaticallyCreateFunctionIdDatabases.java> Received project name: projectName (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> Received project version: projectVersion (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> Received output folder path: C:\Users\malwa\Desktop\fidb (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> Received choice with regards to the usage of a common symbols file: false (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> Collected language IDs and their versions:
      x86:LE:32:default (4.0)
(GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> x86:LE:32:default contains 3 files
(GhidraScript)
INFO Packed database cache: C:\Users\malwa\AppData\Local\ghidra\packed-db-cache (PackedDatabaseCache)
INFO AutomaticallyCreateFunctionIdDatabases.java> [x86:LE:32:default] Generating "projectName (projectVersion)" based on 3 files (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> projectName:projectVersion:x86:LE:32:default (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> 164 total functions visited (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> 42 total functions added (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> 122 total functions excluded (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> Breakdown of exclusions: (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java>   FAILED_FUNCTION_FILTER: 0 (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java>   MEMORY_ACCESS_EXCEPTION: 0 (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java>   IS_THUNK: 0 (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java>   FAILS_MINIMUM_SHORTHASH_LENGTH: 0 (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java>   NO_DEFINED_SYMBOL: 122 (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java>   DUPLICATE_INFO: 0 (GhidraScript)
INFO AutomaticallyCreateFunctionIdDatabases.java> List of unresolved symbols: (GhidraScript)
```


Case study

- Default analysis
- Post FunctionID

```
101 FUN_004e8b20();
102 FUN_004e8b20();
103 FUN_004e88e0();
104 FUN_004e8b20();
105 FUN_004e8880();
106 *param_10 = extraout_R10_01;
107 param_10[1] = extraout_R11_01;
108 param_10[2] = uVar8;
109 param_10[3] = uVar5;
110 FUN_004e8880();
111 FUN_004e88e0();
112 FUN_004e8880();
```

```
101 FUN_004e8b20();
102 FUN_004e8b20();
103 FUN_004e88e0();
104 FUN_004e8b20();
105 p256SubInternal();
106 *param_10 = extraout_R10_01;
107 param_10[1] = extraout_R11_01;
108 param_10[2] = uVar8;
109 param_10[3] = uVar5;
110 p256SubInternal();
111 FUN_004e88e0();
112 p256SubInternal();
```

BSim databases

- Shoehorned
- Automated
- Runs headless

```
<description layout_version="5" major="6" minor="0" settings="0x4d">
<execlist>
<exe>
  <md5>0000a1bccd58d63b924f45dd8ef21ff5</md5>
  <name>curve448-mul-g.obj</name>
  <arch>x86:LE:32:default</arch>
  <compiler>windows</compiler>
  <date millis="0x6">0x65a5a6ba</date>
  <repository>ghidra:/F:/ghidraProjects/libnettle/libnettle</repository>
  <path>libnettle/libnettle_nettle_3.8_release_20220602_msvc15/bin/x86</path>
</exe>
<fdesc name="_nettle_curve448_mul_g" addr="0x2d00">
<lshcosine>
  <hash tf="15">0xbe28207</hash>
  <hash>0x142f8cda</hash>
  <hash>0x2d00064e</hash>
  <hash tf="6">0x3d209d83</hash>
  <hash>0x3fa0950c</hash>
  <hash>0x417e5477</hash>
  <hash tf="6">0x4d88e700</hash>
  <hash>0x4ef74828</hash>
  <hash>0x51f32399</hash>
  <hash tf="26">0x545c6155</hash>
  <hash>0x55123056</hash>
  <hash>0x6c176aad</hash>
  <hash>0x7257ef6c</hash>
  <hash>0x90790e2c</hash>
  <hash tf="9">0x967ef100</hash>
  <hash>0xa150976f</hash>
  <hash tf="7">0xa45c1d72</hash>
  <hash>0xab3b5ale</hash>
  <hash>0xabef06bb</hash>
  <hash>0xc87aca45</hash>
  <hash>0xf0522b88</hash>
</lshcosine>
<call dest="@__security_check_cookie@4">
  <name>unknown</name>
  <compiler></compiler>
</call>
<call dest="__nettle_curve448_eh_to_x">
  <name>unknown</name>
  <compiler></compiler>
</call>
<call dest="__nettle_ecc_mul_g_eh">
  <name>unknown</name>
  <compiler></compiler>
</call>
<call dest="__nettle_gmp_alloc_limbs">
  <name>unknown</name>
  <compiler></compiler>
</call>
```

BSim matches

- Provide context
- Identical matches
 - Renamed as-is

Decompile: crypto/elliptic.p256PointDoubleAsm - (qbit_258f9a7bab553b4731da8913be33134ec29fe3..

```
1
2 /* Old name: FUN_004e9ba0
3
4     Detected name: "crypto/elliptic.p256PointDoubleAsm"
5
6     Original file, similarity, significance, architecture
7     emptyProject_gol.9rc1_windows_amd64  1  251.462  x86:LE:64:default
8     emptyProject_gol.9.2rc2_windows_amd64  1  251.462  x86:LE:64:default
9     emptyProject_gol.9.2_windows_amd64  1  251.462  x86:LE:64:default
10    emptyProject_gol.16betal_windows_amd64  1  251.462  x86:LE:64:default
11    emptyProject_gol.15betal_windows_amd64  1  251.462  x86:LE:64:default
12    emptyProject_gol.15.5_windows_amd64  1  251.462  x86:LE:64:default
13    emptyProject_gol.15.15_windows_amd64  1  251.462  x86:LE:64:default
14    emptyProject_gol.15.14_windows_amd64  1  251.462  x86:LE:64:default
15    emptyProject_gol.14betal_windows_amd64  1  251.462  x86:LE:64:default
16    emptyProject_gol.14.3_windows_amd64  1  251.462  x86:LE:64:default
17    emptyProject_gol.14.14_windows_amd64  1  251.462  x86:LE:64:default
18    emptyProject_gol.13rc1_windows_amd64  1  251.462  x86:LE:64:default
19    emptyProject_gol.13.7_windows_amd64  1  251.462  x86:LE:64:default
20    emptyProject_gol.13.15_windows_amd64  1  251.462  x86:LE:64:default
21    emptyProject_gol.12rc1_windows_amd64  1  251.462  x86:LE:64:default
22    emptyProject_gol.12.9_windows_amd64  1  251.462  x86:LE:64:default
23    emptyProject_gol.12.8_windows_amd64  1  251.462  x86:LE:64:default
24    emptyProject_gol.12.7_windows_amd64  1  251.462  x86:LE:64:default
25    emptyProject_gol.12.2_windows_amd64  1  251.462  x86:LE:64:default
26    emptyProject_gol.12.13_windows_amd64  1  251.462  x86:LE:64:default */
27
28 void crypto/elliptic.p256PointDoubleAsm
```

BSim matches

- Provide context
- Generic matches
 - Renamed with a prefix

```
Decompile: golang_FUN_004d0b60 - (qbit_258f9a7bab553b4731da8913be33134ec29fe3061f... Ro
1
2 /* Old name: golang_FUN_004d0b60
3
4     Old name: FUN_004d0b60
5
6     Detected name: "crypto/internal/nistec/ fiat::crypto/internal/nistec/ fiat.p384Add"
7
8     Original file, similarity, significance, architecture
9     emptyProject_gol.19_windows_amd64  0.856  134.93  x86:LE:64:default
10    emptyProject_gol.19.8_windows_amd64  0.856  134.93  x86:LE:64:default
11    emptyProject_gol.19.6_windows_amd64  0.856  134.93  x86:LE:64:default
12    emptyProject_gol.19.5_windows_amd64  0.856  134.93  x86:LE:64:default
13    emptyProject_gol.19.4_windows_amd64  0.856  134.93  x86:LE:64:default
14    emptyProject_gol.19.3_windows_amd64  0.856  134.93  x86:LE:64:default
15    emptyProject_gol.19.2_windows_amd64  0.856  134.93  x86:LE:64:default
16    emptyProject_gol.19.1_windows_amd64  0.856  134.93  x86:LE:64:default
17    emptyProject_gol.19.13_windows_amd64 0.856  134.93  x86:LE:64:default
18    emptyProject_gol.19.12_windows_amd64 0.856  134.93  x86:LE:64:default
19    emptyProject_gol.19.11_windows_amd64 0.856  134.93  x86:LE:64:default
20    emptyProject_gol.19.10_windows_amd64 0.856  134.93  x86:LE:64:default
21
22    Detected name: "crypto/elliptic/internal/ fiat::crypto/elliptic/internal/ fiat.p384Add"
23
24    Original file, similarity, significance, architecture
25    emptyProject_gol.18_windows_amd64  0.856  134.93  x86:LE:64:default
26    emptyProject_gol.18.9_windows_amd64 0.856  134.93  x86:LE:64:default
27    emptyProject_gol.18.8_windows_amd64 0.856  134.93  x86:LE:64:default
28    emptyProject_gol.18.6_windows_amd64 0.856  134.93  x86:LE:64:default
29    emptyProject_gol.18.5_windows_amd64 0.856  134.93  x86:LE:64:default
30    emptyProject_gol.18.3_windows_amd64 0.856  134.93  x86:LE:64:default
31    emptyProject_gol.18.2_windows_amd64 0.856  134.93  x86:LE:64:default
32    emptyProject_gol.18.10_windows_amd64 0.856  134.93  x86:LE:64:default */
33
34 void golang_FUN_004d0b60(uint *param_1,uint *param_2,uint *param_3,undefined param_4,
```

Case study

- Post FunctionID
- Post BSim

```
101 FUN_004e8b20();
102 FUN_004e8b20();
103 FUN_004e88e0();
104 FUN_004e8b20();
105 p256SubInternal();
106 *param_10 = extraout_R10_01;
107 param_10[1] = extraout_R11_01;
108 param_10[2] = uVar8;
109 param_10[3] = uVar5;
110 p256SubInternal();
111 FUN_004e88e0();
112 p256SubInternal();
```

```
128 golang_FUN_004e8b20();
129 golang_FUN_004e8b20();
130 golang_FUN_004e88e0();
131 golang_FUN_004e8b20();
132 p256SubInternal();
133 *param_10 = extraout_R10_01;
134 param_10[1] = extraout_R11_01;
135 param_10[2] = uVar8;
136 param_10[3] = uVar5;
137 p256SubInternal();
138 golang_FUN_004e88e0();
139 p256SubInternal();
```

Bulk

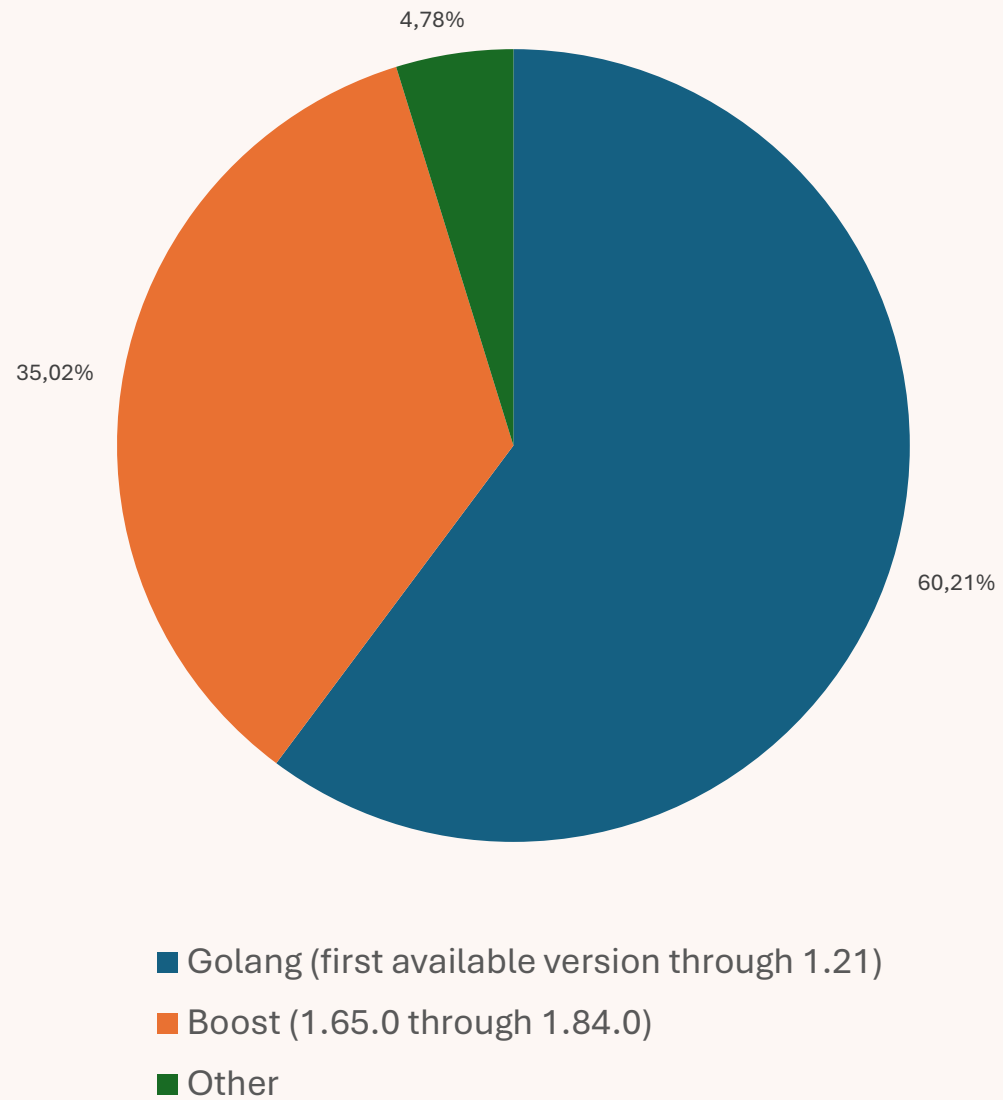


Hundreds of gigabytes of benign binaries



117 GiB of BSim signatures

Bulk FunctionID signature generation





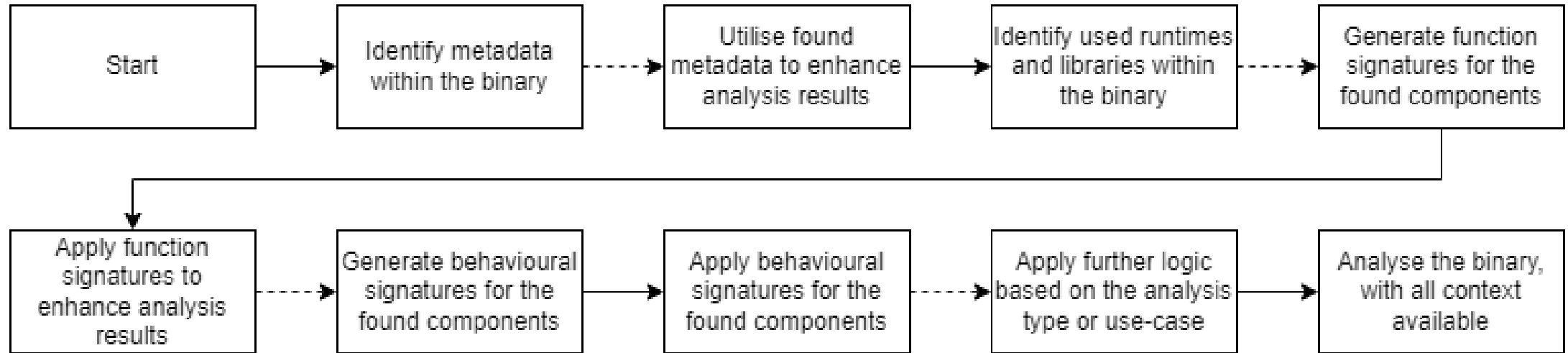
Bulk signature generation

Project name	FunctionID signatures
Golang (first available version through 1.21)	48,472,262
Boost (1.65.0 through 1.84.0)	28,191,439
libxml2	1,608,164
openssl	818,210
libgcrypt	673,280
libnettle	447,443
liblzma	108,279
msvc-empty	86,479
uclibc	47,394
libzlib	16,292
libsodium	16,093
avr-libc	15,616
libbzip2	7,433

Case study: qBit stealer

Step	Recovered symbols
Default analysis	1/2016 (0.05%)
FunctionID recovery	850/2015 (42.2%)
Bsim recovery	319/2015 (15.8%)
<u>Total</u>	<u>1169/2015 (58.0%)</u>

```
import (  
    "flag"  
    "fmt"  
    "github.com/gizak/termui/v3"  
    "github.com/gizak/termui/v3/widgets"  
    "log"  
    "os"  
    "sync"  
    "time"  
)
```



Legend

Unconditional flow →

Conditional flow - - ->



Questions?

- info@maxkersten.nl
- [/in/Libranalysis](#) on LinkedIn
- [@Libranalysis](#) on Twitter
- <https://github.com/advanced-threat-research/>
 - [GhidraScripts](#)
 - [FIDBs](#)
 - [BSim](#)
 - [Golang-Runtime-Binaries](#)